

Indian Institute Of Information Technology Guwahati



Project Report 1

Group Number - 14

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Section: CS21

Branch: B.Tech 2nd Year CSE

Course: DBMS Lab

Code: CS241

ER Diagram Report

ER Diagram design for a Real Estate office

System Interfaces:

- Admin Interface
- Office Interface
- Agent Interface
- User Interface

Assumed Entities:

1. User - User is general entity that can act as any of the roles namely Buyer/Seller/Land Lord/Tenant. We did not associate a tag with the user, because he can, at the same time be a seller of some property while buyer of some other.
2. Property - Property is any asset owned by some user, which he/she wants to sell. It can be on rent or sell
3. Agent - An agent is an entity who act as a mediator between Buyers and Sellers.
4. Transaction - A Transaction is the central entity that records the successful fulfilment of a deal between a User, an Agent, and a Property.
5. ListingToken - It contains the information about property in intermediate phase
6. Notification - It contains all the inbox messages of both the agents and users
7. Performance - It records the performance data of an Agent
8. Images - Contain the preview of the property
9. Office - Manages all the Agents and track Property

Assumed Relations:

1. Owns - Property to User(Owner) | Many to One
2. Generates - ListingToken to User(Owner) | Many to One
3. Makes - Transaction to user(Buyer) | Many to One
4. Of - ListingToken to Property | Many to One
5. Deal - ListingToken to Transaction | One to One
6. Sends - Notification to User | Many to One
7. Receives - Notification to User | Many to One
8. Preview - Images to Property | Many to One
9. Handles - Agent to Transaction | One to Many
10. Tracking - Performance to Agent | One to One
11. Accepts - Agent to ListingToken | One to Many
12. Records - Office to Property | One to Many
13. Manages - Office to Agents | One to Many
14. Is - Agent to User | One to One

Assumed Rules:

1. Every property must have an owner among the users, an owner can own multiple properties.
2. A user(Owner) enters the details of property and generates and Listing token to advertises it to agents, this property(Without Agent) will be visible to all agents in the Agent's Interface
3. Agents will search through many such properties and accept invite for properties of his choice.
4. Once agent accepts, he will be associated to that ListingToken or Property, and that property will be shown to the Buyers in the users Interface.
5. A user who wish to buy/rent property will search and contact the associated agent.
6. Agent will lead towards a transaction, and property will be sold/rented.
7. The Transaction table will maintain the record of the buyer, property, agent, and Previous owner.
8. After the transaction if the property is sold then the ownership is transferred.
9. Listing token is treated as a task, once task is complete, the token is closed, and status is changed to sold. The property can be transferred from one buyer to other buyer later.
10. If the buyer again want to sell the property, then he will again generate a token, for the transferred property.

Individual Contribution

1. Anmol Kumar (2401031)

Discussed the required entities and their attributes for the initial ER Diagram, Tested the diagram for multiple queries

2. Anmol Kumar (2401032)

Designed initial phase of ER diagram on paper and discussed the flow of the system design, considering all the future requirements

3. Ansik Singh Tomar (2401037)

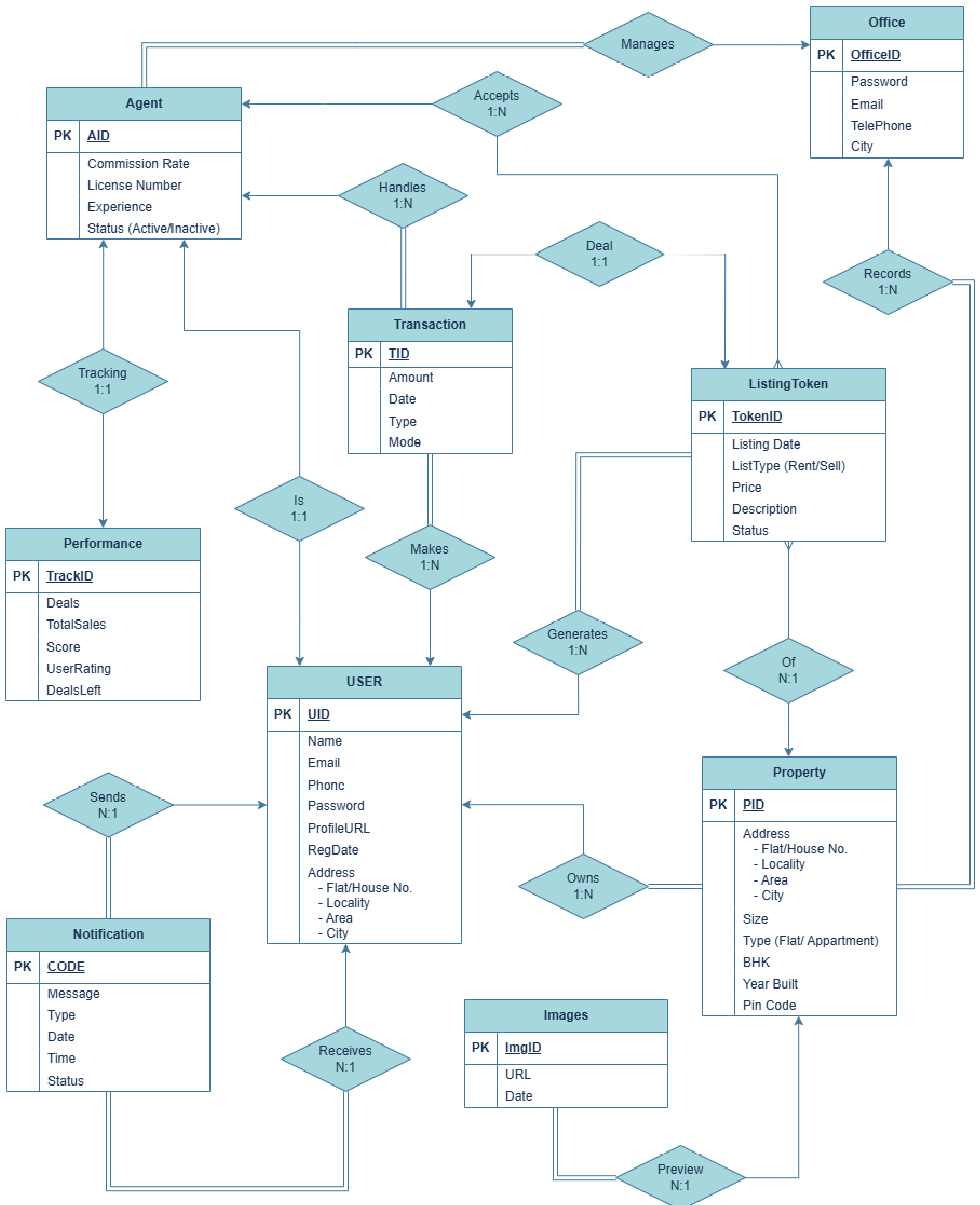
Drawn the final ER diagram using the online tool depicting all the relations and entities with cardinalities, and drafted this report.

4. Anuved Pratap Singh (2401041)

Designed the prefinal phase of ER diagram on paper, discussed the constraints, helped in assuming the rules defining the Entities

All the team members we involved in discussions, and held several meeting.

ER Diagram



Schema

Performance	
PK	<u>TrackID</u>
FK	AID Deals TotalSales Score UserRating DealsLeft

Notification	
PK	<u>CODE</u>
FK	SID
FK	RID
	Message Type Date Time Status

ListingToken	
PK	<u>TokenID</u>
FK	PID
FK	AID Listing Date ListType (Rent/Sell) Price Description Status

Agent	
PK - FK	<u>AID (=UID)</u>
FK	OfficeID Commission Rate License Number Experience Status (Active/Inactive)

Property	
PK	<u>PID</u>
FK	OwnerID
FK	OfficeID
	HouseNo Locality Area City Pin Size Type (Flat/ Apartment) BHK Year Built

Images	
PK	<u>ImgID</u>
FK	PID URL Date

Office	
PK	<u>OfficeID</u>
	Password Email TelePhone City

Transaction	
PK	<u>TID</u>
FK	AID
FK	TokenID
FK	BuyerID
	Amount Date Type Mode

USER	
PK	<u>UID</u>
	Name Email Phone Password ProfileURL RegDate Area City Pin